

Module-4 (Assignment Day 1)

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.



Three real-life scenarios where a multimeter is used:

1. **Computer not powering on:** A multimeter checks the PSU output voltage to confirm whether the power supply is working correctly.
2. **Laptop battery not charging:** A multimeter measures the battery or charger voltage to identify a faulty battery or adapter.
3. **Smartphone not charging:** A multimeter checks the charging port and battery voltage to find power or connection problems.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.



The POST card displays hexadecimal POST codes while the motherboard checks different hardware during startup.

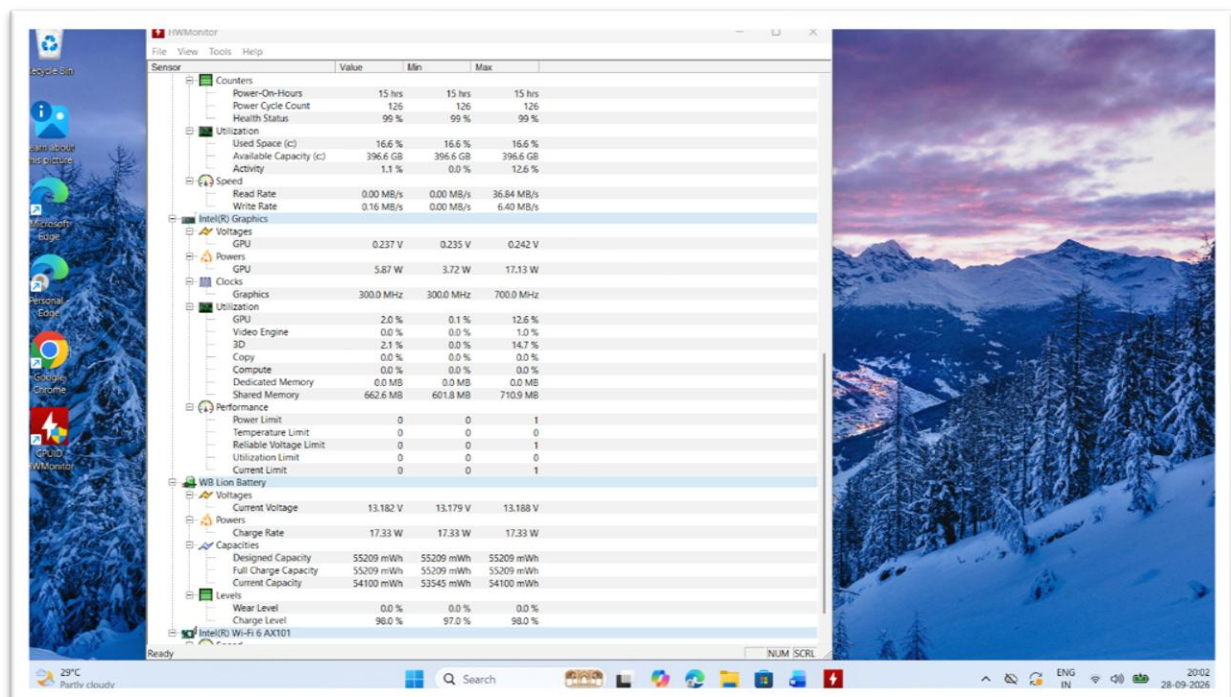
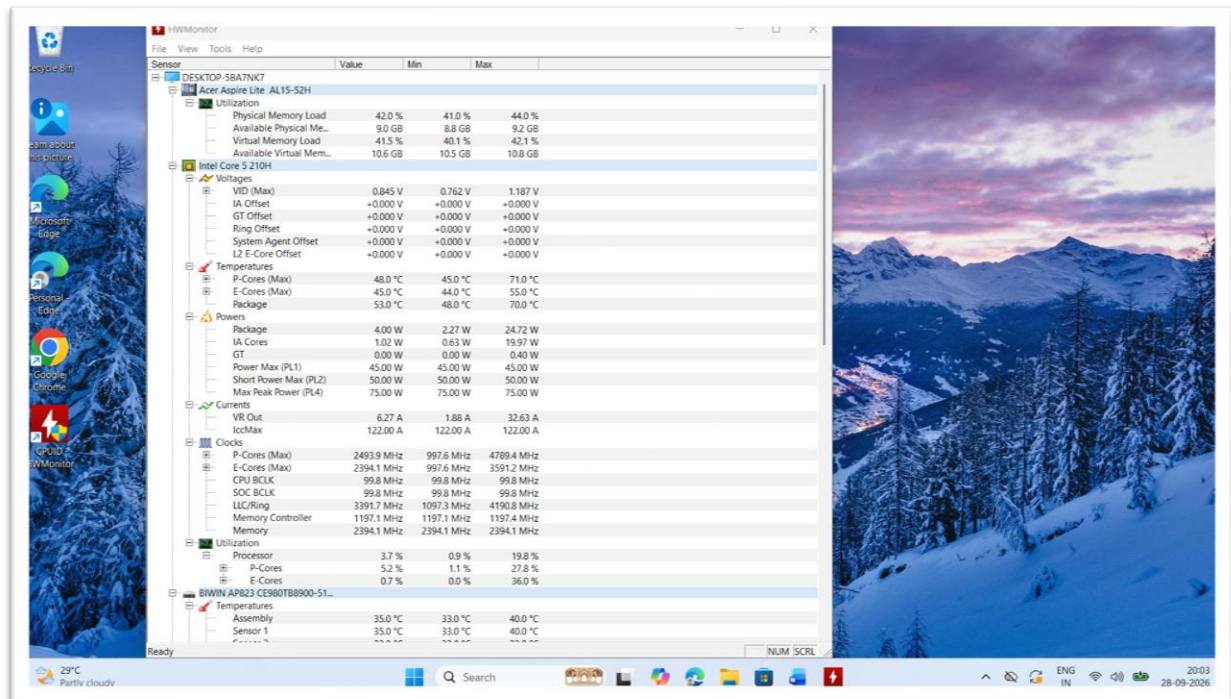
In the demo, the codes help show which stage of the boot process the computer has reached.

When the code stops or shows an error, it indicates a possible RAM, CPU, graphics, or motherboard problem.

The POST card therefore helps identify the faulty hardware when the computer does not boot or has no display.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.



4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.



Diagnostic Tools Comparison

Tool	What it checks	Example Problem Detected
Multimeter	Electrical characteristics such as voltage, current, and resistance in hardware components, power supply rails, and circuits at a physical level.	A failing power supply unit (PSU) delivering out-of-spec voltage
POST Card	Power-On Self-Test (POST) status codes generated by the motherboard BIOS/UEFI during the boot process before the operating system loads.	A motherboard halting during boot due to a RAM initialization failure, outputting a specific diagnostic hex code
Software Diagnostics	Operating system-level hardware metrics, including component temperatures, S.M.A.R.T. drive health, clock speeds, and resource utilization.	An overheating CPU thermal throttling due to dried-out thermal paste, visible as excessively high core temperatures in monitoring software.